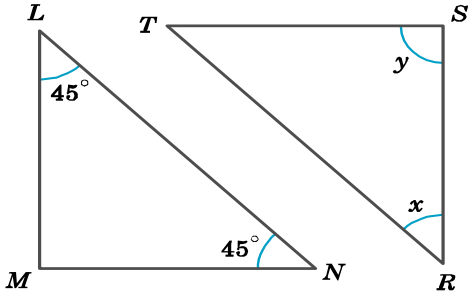
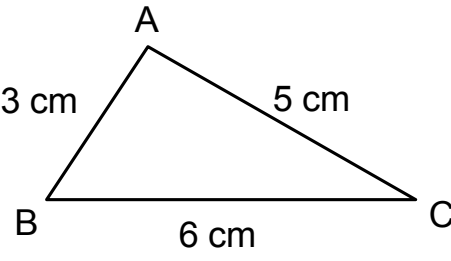
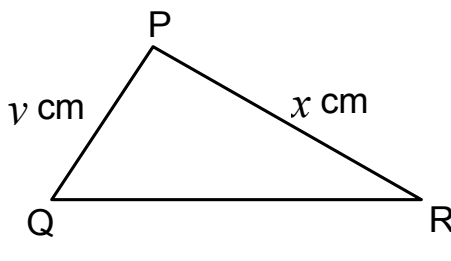


姓名：_____

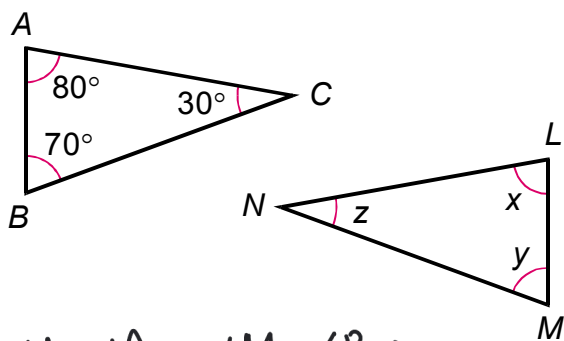
班別：_____ ()

G11.1 全等三角形定義

1. 根據下列條件，求下列各圖中的未知量。

例一 $\triangle RST \cong \triangle LMN$ 	$\therefore \angle R = \angle L$ $\therefore x = 45^\circ$ $\therefore \angle S = \angle M$ $\therefore y = 180^\circ - 45^\circ - 45^\circ$ $\underline{y = 90^\circ}$	學習影片/例題 	Gmath 全等三角形命名 
例二 $\triangle PQR \cong \triangle ABC$ $\therefore PQ = AB$ $\underline{\therefore v = 3}$ $\therefore PR = AC$ $\underline{\therefore x = 5}$  			

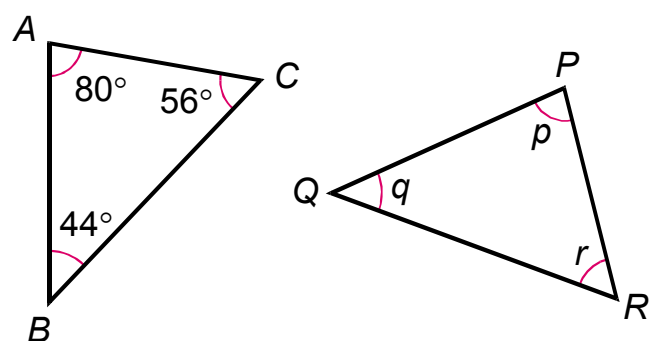
(a) $\triangle ABC \cong \triangle LMN$



$\angle L = \angle A$; $\angle M = \angle B$;
 $x = 80^\circ$, $y = 70^\circ$

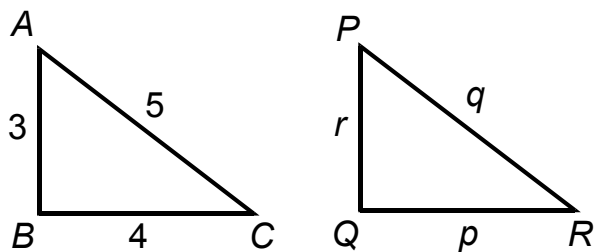
$\angle N = \angle C$
 $z = 30^\circ$

(b) $\triangle ABC \cong \triangle PQR$



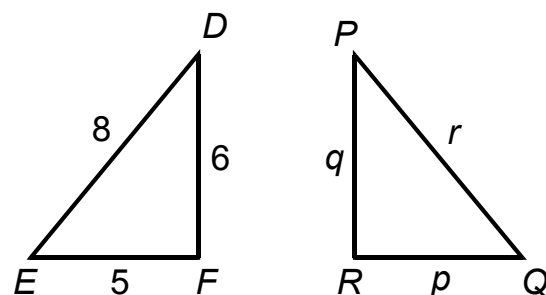
$\angle P = \angle A$; $\angle Q = \angle B$; $\angle R = \angle C$
 $p = 80^\circ$, $q = 44^\circ$, $r = 56^\circ$

(c) $\triangle ABC \cong \triangle PQR$



$QR = BC$; $PR = AC$; $PQ = AB$
 $p = 4$, $q = 5$, $r = 3$

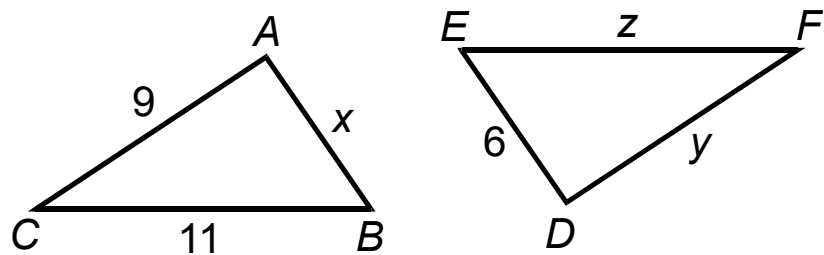
(d) $\triangle DEF \cong \triangle PQR$



$QR = EF$; $PR = DF$; $PQ = DE$
 $p = 5$, $q = 6$, $r = 8$

2. 根據下列條件，求下列各圖中的未知量。

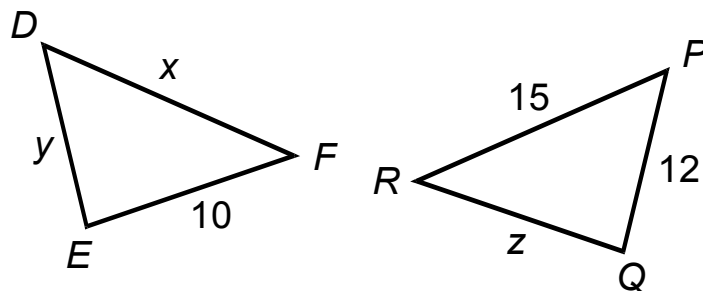
(a) $\triangle ABC \cong \triangle DEF$



$$AB = DE ; FD = CA ; EF = BC$$

$$x = 6'' \quad y = 9'' \quad z = 11''$$

(b) $\triangle DEF \cong \triangle PQR$



$$DF = PR ; DE = PQ ; RQ = FE$$

$$x = 15'' \quad y = 12'' \quad z = 10''$$

挑戰題

3. 圖中， $\triangle ABE \cong \triangle CBD$

(a) 求 w 。

$$BD = BE$$

$$w = 5''$$

(b) 求 x 。

$$\angle ABE = \angle CBD$$

$$x = 2x - 40^\circ$$

$$x - 2x = -40^\circ$$

$$-x = -40^\circ$$

$$x = 40''$$

(c) 求 y 和 z 。

$$\angle DCB = \angle EAB$$

$$y = 60''$$

$$z + x + 60^\circ = 180^\circ \text{ (三角形內角和)}$$

$$z + 40^\circ + 60^\circ = 180^\circ$$

$$z = 180^\circ - 40^\circ - 60^\circ$$

$$z = 80''$$

